Chapter5 Light

Lesson 1 Light Ray

- Sources of light
- Rays and beams
- Shadows
- Speed of light
- Practical work: The pinhole camera

1. Sources of light

You can see an object only if light from it enters your eyes. Some objects such as the Sun, electric lamps and candles make their own light. We call these **luminous** sources.

Most things you see do not make their own light but reflect it from a luminous source. They are **non-luminous** objects. This page, you and the Moon are examples. Figure 26.1 shows some others.

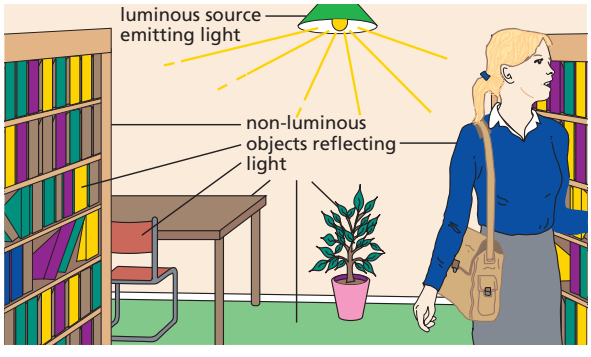


Figure 26.1 Luminous and non-luminous objects

Luminous sources radiate light when their atoms become 'excited' as a result of receiving energy. In a light bulb, for example, the energy comes from electricity. The 'excited' atoms give off their light haphazardly in most luminous sources.

A light source that works differently is the **laser**, invented in 1960. In laser light sources the excited atoms act together and emit a narrow, very bright beam of light. The laser has a host of applications. It is used in industry to cut through plate metal, in scanners to read bar codes at shop-and library check-outs, in CD players, in optical fibre telecommunication systems, in delicate medical operations on the eye or inner ear (for example Figure 26.2), in printing and in surveying and range-finding.



Figure 26.2 Laser surgery in the inner ear

2. Rays and beams

Sunbeams streaming through trees (Figure 26.3) and light from a cinema projector on its way to the screen both suggest that light travels in straight lines. The beams are visible because dust particles in the air reflect light into our eyes.

The direction of the path in which light is travelling is called a **ray** and is represented in diagrams by a straight line with an arrow on it. A beam is a stream of light and is shown by a number of rays, as in Figure 26.4. A beam may be **parallel**, **diverging** (spreading out) or **converging** (getting narrower).

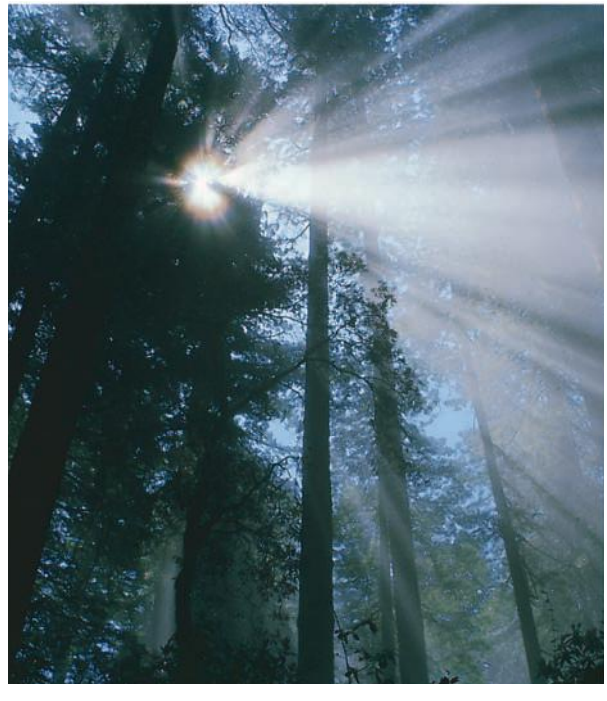


Figure 26.3 Light travels in straight lines

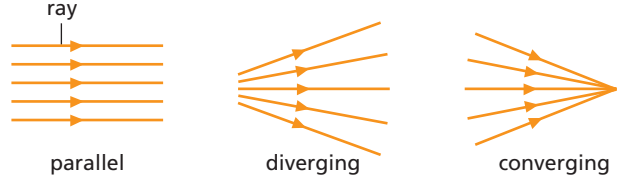


Figure 26.4 Beams of light

Practical work

The pinhole camera

A simple pinhole camera is shown in Figure 26.5a. Make a small pinhole in the centre of the black paper. Half darken the room. Hold the box at arm's length so that the pinhole end is nearer to and about 1 metre from a luminous object, such as a carbon filament lamp or a candle. Look at the **image** on the screen (an image is a likeness of an object and need not be an exact copy).

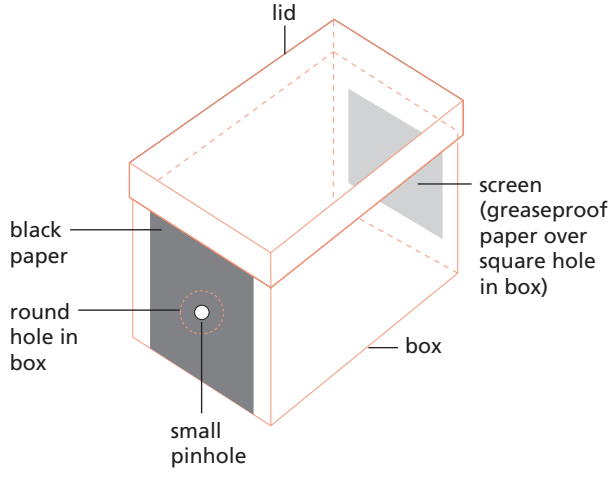
Can you see *three* ways in which the image differs from the object? What is the effect of moving the camera closer to the object?

Make the pinhole larger. What happens to the

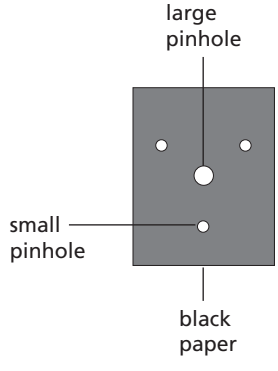
(i) brightness,
(ii) sharpness,
(iii) size of the image?

Make several small pinholes round the large hole (Figure 26.5b), and view the image again.

The formation of an image is shown in Figure 26.6.



a A pinhole camera



b

Figure 26.5

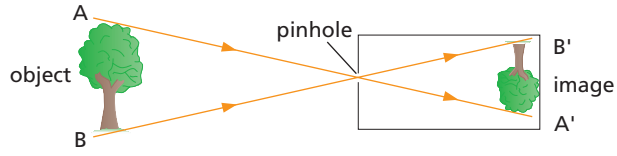
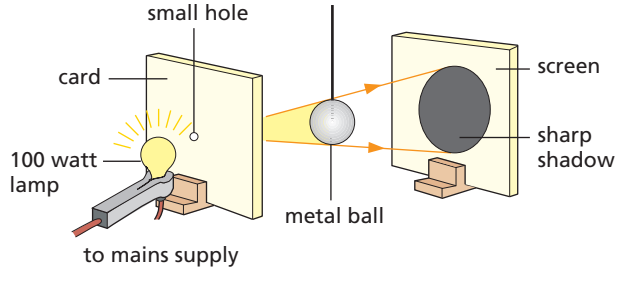


Figure 26.6 Forming an image in a pinhole camera

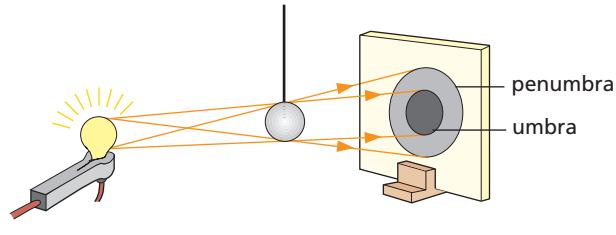
3. Shadows

Shadows are formed for two reasons. First, because some objects, which are said to be **opaque**, do not allow light to pass through them. Secondly, light travels in straight lines.

The sharpness of the shadow depends on the size of the light source. A very small source of light, called a **point source**, gives a sharp shadow which is equally dark all over. This may be shown as in Figure 26.7a where the small hole in the card acts as a point source.



a With a point source



b With an extended source

Figure 26.7 Forming a shadow

If the card is removed the lamp acts as a large or **extended source** (Figure 26.7b). The shadow is then larger and has a central dark region, the **umbra**, surrounded by a ring of partial shadow, the **penumbra**. You can see by the rays that some light reaches the penumbra but none reaches the umbra.

4 Speed of light

Proof that light travels very much faster than sound is provided by a thunderstorm. The flashoflightning is seen before the thunder is heard. The length of the time lapse is greater the further the observer is from the storm.

The speed of light has a defi nite value; light does not travel instantaneously from one point to another but takes a certain, very small time. Its speed is about 1 million times greater than that of sound.That is about 300,000km/s.

Checklist

After studying this chapter you should be able to

- give examples of effects which show that light travels in a straight line,
- explain the operation of a pinhole camera and draw ray diagrams to show the result of varying the object distance or the length of the camera,
- draw diagrams to show how shadows are formed using point and extended sources, and use the terms umbra and penumbra,
- recall that light travels much faster than sound.

Questions

1 How would the size and brightness of the image formed by a pinhole camera change if the camera were made longer?

2 What changes would occur in the image if the single pinhole in a camera were replaced by
- **a** four pinholes close together,
- **b** a hole 1 cm wide?

3 In Figure 26.8 the completely dark region on the screen is
A PQ **B** PR **C** QR **D** QS **E** RS

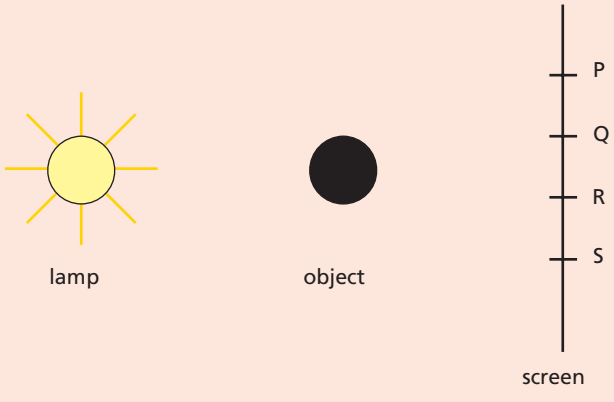


Figure 26.8

4 When watching a distant firework display do you see the cascade of lights before or after hearing the associated bang? Explain your answer.